HPC Cluster Tutorial:
Module System and Slurm Commands

Aatiz Ghimire

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1 Introduction

This tutorial covers the **Environment Modules / Lmod** system and **Slurm** workload manager commonly used on HPC clusters (including TU Super Computing Center).

2 1. Module System (Lmod / Environment Modules)

2.1 Basic Commands

```
# List all available modules
module avail

# Search for packages
module spider <keyword>
module spider gcc/12.3.0

# Currently loaded modules
module list

# Load / Unload
module load gcc/12.3.0
module unload openmpi/4.1.6

# Swap versions
module swap gcc/11.4.0 gcc/12.3.0

# Clean everything
module purge

# Save / Restore collections
module save my-dl-env
module restore my-dl-env
```

2.2 Common Loading Patterns

```
module load gcc/12.3.0 openmpi/4.1.6
module load cuda/12.4 cudnn/9.1
module load gromacs/2024.2
module load vasp/6.4.3
module load miniforge3/24.3
```

Add to ~/.bashrc for convenience:

```bash
if [ -f /etc/profile.d/modules.sh ]; then
    . /etc/profile.d/modules.sh
fi
module restore default
```

3 2. Slurm Workload Manager

3.1 Cluster Information

```bash
sinfo -N -l                    # Node status
sinfo -s                       # Partition summary
scontrol show partition
```

3.2 Job Monitoring

```bash
squeue -u $USER                # Your jobs
squeue -o "%.18i %.9P %.8j %.8u %.2t %.10M %.6D %R"

scontrol show job <jobid>
sacct -u $USER -S 2026-07-01
```

3.3 Interactive Jobs

```bash
srun --partition=gpu --gres=gpu:1 --cpus-per-task=4 --mem=16G \
     --time=01:00:00 --pty bash -i
```

3.4 Batch Job Script (`job.sh`)

```bash
#!/bin/bash
#SBATCH --job-name=my_job
#SBATCH --partition=gpu
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=8
#SBATCH --gres=gpu:1
#SBATCH --mem=32G
#SBATCH --time=24:00:00
#SBATCH --output=%x_%j.out
#SBATCH --error=%x_%j.err
#SBATCH --mail-type=END,FAIL
#SBATCH --mail-user=aatizghm@gmail.com

module purge
module load gcc/12.3.0 openmpi/4.1.6 cuda/12.4
module load gromacs/2024.2

cd /scratch/$USER/my_simulation
srun gmx mdrun -deffnm md -v
```

Submit with:

```bash
sbatch job.sh
scancel <jobid>
```

3.5 Useful Slurm Commands

- `salloc` — allocate resources interactively
- `--array=1-100%10` — job arrays
- `sprio -u $USER` — priority information
- `sreport cluster AccountUtilizationByUser` — usage report

4 Best Practices

- Always `module purge` before loading new modules in scripts
- Use `/scratch` for large I/O
- Test with interactive `srun` before submitting batch jobs
- Save common module collections with `module save`